

Derivative

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1 Limits

To calculate the derivative, we look at the change over time h ,

$$\frac{f(x+h) - f(x)}{h} \quad h \neq 0$$

So for the function $f(x) = x^2$,

$$\begin{aligned} f'(x) &= \frac{(x+h)^2 - x^2}{h} \\ &= \frac{\cancel{x^2} + 2xh + h^2 - \cancel{x^2}}{h} \\ &= \frac{2xh + h^2}{h} \\ &= \frac{\cancel{h}(2x + h)}{\cancel{h}} \\ &= 2x + h \end{aligned}$$

$$\lim_{h \rightarrow 0} f'(x) = 2x$$

For function $f(x) = \sqrt[2]{x}$,

$$\begin{aligned}
 f'(x) &= \frac{\sqrt[2]{x+h} - \sqrt[2]{x}}{h} \\
 &= \frac{\sqrt[2]{x+h} - \sqrt[2]{x}}{h} \cdot \frac{\sqrt[2]{x+h} + \sqrt[2]{x}}{\sqrt[2]{x+h} + \sqrt[2]{x}} \\
 &= \frac{\cancel{x} + h - \cancel{x}}{h(\sqrt[2]{x+h} + \sqrt[2]{x})} \\
 &= \frac{h}{h(\sqrt[2]{x+h} + \sqrt[2]{x})} \\
 &= \frac{1}{\sqrt[2]{x+h} + \sqrt[2]{x}} \\
 \lim_{h \rightarrow 0} f'(x) &= \frac{1}{2\sqrt[2]{x}}
 \end{aligned}$$